Sprinkler Actuator Power Budget

Team Number:	101
Project Name:	**Projectz**
Team Member Names:	**Liam Mabbutt**
Version:	**1**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,

All Major Components	Component Name	Part Number	yVoltageRan	#	aximumCurre	alCurrent(mA)	Unit
	9V DC Motor	1528-1150-ND	3-9V	1	500	500	mA
	FAN8100N	2156-FAN8100	1.8-9	1	1500	1500	mA
	PIC Curiosity Nano Microcon	DM164142-PIC1	4.4-5.25	1	15	15	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in

9V Power Rail	Component Name	Part Number	yVoltageRan	#	aximumCurre	alCurrent(mA)	Unit
	9V DC Motor	1528-1150-ND	3-9V	1	500	500	mA
	FAN8100N	2156-FAN8100	1.8-9	1	1500	1500	mA
						0	mA
						0	mA
						0	mA
					Subtotal	2000	mA
					Safety Margin	25%	
					Total Current Required on +9V Rail	2500	mA
c1. Regulator or Source Ch	9V 3A Wall Source		9V	1	3000	3000	mA
					Total Remaining Current Available on +9V Rail	500	mA

+5V Power Rail	Component Name	Part Number	yVoltageRan	#	aximumCurre	alCurrent(mA)	Unit
	PIC Curiosity Nano Microcon	DM164142-PIC1	4.4-5.25	1	15	15	mA
						0	mA
						0	mA
						0	mA
						0	mA
					Subtotal	15	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	18.75	mA
c2. Regulator or Source Ch	+5V Regulator	LM7805	5V	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	1481.25	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component							
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the							
External Power Source 1	**Component Name**	**Part Number**	**yVoltageRan**	**Outpu**	**aximumCurre**	**alCurrent(mA)**	**Unit**
Power Source 1 Selection	Plug-in Wall Supply	BestCH	240	9V	1.5A	1500	mA
Power Rails Connected to External Power Source 1	+5V Regulator	LM7805	35V	5V	1.5A	1500	mA
							mA
	Total Remaining Current Available on External Power Source 1					0	mA

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V

If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit